

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	CSA07	Course Title:	Computer Networks
CO Assessed:	CO5 – K3 - Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	10% of CIA	Slot	A
CO Statement:	Measure network performance using key metrics with simulation tools (BL4, BL5)		
Student Name:	K.Dinesh	Reg. No.:	1921260051
Section / Batch:		Date:	

Code of Conduct

I K.Dinesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Duration: 45 minutes. In-Class Assessment
- Complete all four tasks individually.
- Start Wireshark before generating the network activity.
- Select the active network interface (Wi-Fi/Ethernet) and start packet capture.
- Apply the given constraint strictly. You must not use alternative filters or tools to obtain the answer.
- Record the required Wireshark display filter and identify the required packet/protocol/port.
- Submit the GitHub folder/repository link in Google Classroom (GCR).

Section / Topic	Focus	Q. No.
13. Application Layer Protocols	DNS – Domain Name Resolution	Q1
13. Application Layer Protocols	HTTP/HTTPS – Web Communication	Q2
13. Application Layer Protocols	FTP – File Transfer and Control Connection	Q3
13. Application Layer Protocols	SMTP – Electronic Mail Transmission	Q4

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
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Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annex: 3. Constraint-Based Problem Solving

1. DNS Resolution – Google

Open the website:

Google: <https://www.google.com/?zx=1787560655258> Start

Wireshark before opening the browser.

Constraint

You are allowed to inspect **only DNS packets**.

Task

Identify the packet containing the DNS response that resolves `www.google.com`.

Digital response

Select the correct packet number from the options.

Wireshark filter: **DNS**

2. HTTPS Web Access – Wikipedia

Clear the browser cache, start Wireshark, and open:

Wikipedia: <https://www.wikipedia.org/>

Constraint

You may use only **one display filter** and cannot inspect browser developer tools.

Task

Identify the transport-layer protocol carrying the encrypted web connection. [Options](#)

- TCP
- UDP
- ICMP
- ARP

Correct answer: **TCP** .

3. FTP Traffic Challenge – Real FTP Test Service

Use a public FTP test service provided for testing: Rebex

FTP Test Server: <https://test.rebex.net/>

Capture the connection using an FTP client.

Constraint

You may use only one port-based filter.

Task

Identify the FTP control connection. [Options](#)

- Port 21
- Port 25
- Port 53
- Port 161

Wireshark filter: **tcp.port == 21** .

4. SMTP Traffic Identification – Mailtrap Sandbox

Use an SMTP testing environment such as:

Mailtrap Email Sandbox: <https://mailtrap.io/email-sandbox/>

Configure the provided sandbox SMTP credentials in a mail-testing client or application and capture the connection.

Constraint

Students can inspect only packets belonging to the configured SMTP connection.

Task

Identify the application protocol responsible for transmitting the outgoing email.

Options

- SMTP
- POP • DNS
- SNMP

Correct response: **SMTP** .